From the structure of $\boldsymbol{W}_V^*$ (Lemma 9), we observe that the predicted logit $\boldsymbol{W}_V \boldsymbol{X}_{1:T^*} \mathrm{Softmax}(\boldsymbol{X}_{1:T^*}^\top \boldsymbol{W}_{KQ} \boldsymbol{x}_{T^*})$ has a peak on the token u, meaning that OOD generalization fails. □

Remark 3. *Here we worked on the ratio between $q_{\ell^*} T(\ell^*)^{-1} = \max_\ell (2\ell+3)^{-1} q_\ell$ and $\mathbb{E}[T(\ell)^{-1}] = \sum_\ell (2\ell+3)^{-1} q_\ell$. We can immediately show (using Assumption 2) $\frac{1}{3}\max_\ell (\ell)^{-1} q_\ell \le \max_\ell (2\ell+3)^{-1} q_\ell \le \frac{1}{2}\max_\ell (\ell)^{-1} q_\ell$ and $\frac{1}{3}\sum_\ell (\ell)^{-1} q_\ell \le \sum_\ell (2\ell+3)^{-1} q_\ell \le \frac{1}{2}\sum_\ell (\ell)^{-1} q_\ell$, then we do not distinguish these two definitions of max-sum ratio.*

Similarly we can show the following upper bound:

Lemma 15. *If it holds that*

$$\frac{\max_\ell q_\ell \ell^{-1}}{\sum_\ell q_\ell \ell^{-1}} \ge \epsilon(N_{\mathrm{trg}})$$

where $\epsilon(N_{\mathrm{trg}}) = \Theta(N_{\mathrm{trg}}^{-1})$, the pretrained transformer via Algorithm 1 can generalize OOD.

Proof. In the same vain as the proof of the lower bound, it suffices to show $s_{\ell_1^*+2} \ge 2 s_t$ for any $t \ne \ell_1^* + 2$, going the other way around Lemma 13.

First we have

$$\begin{aligned}
s_{\ell_1^*+2} &\ge \tilde{\eta} \frac{q^*}{N_{\mathrm{trg}}} \sum_{w=1}^{N_{\mathrm{trg}}} [-2(T^*)^{-2} \mathbb{I}(\ell_1^* + 2 \le T^*)] \\
&\quad + \frac{\tilde{\eta}}{N_{\mathrm{trg}}} [((2\ell_1^*+3)^{-1} + 2(2\ell_1^*+3)^{-2}) q_{\ell_1^*} + ((2\ell_1^*+1)^{-1} + 2(2\ell_1^*+1)^{-2}) q_{\ell_1^*-1} \\
&\qquad - 2\mathbb{E}[T^{-2}] + \mathbb{E}[T^{-1}]] + O_\infty(\tilde{\eta} N_{\mathrm{trg}} \cdot N^{-1}) \\
&\ge \frac{\tilde{\eta}}{N_{\mathrm{trg}}} \mathbb{E}[T^{-1}] - 2\frac{\tilde{\eta}}{N_{\mathrm{trg}}} \mathbb{E}[T^{-2}] - 2\tilde{\eta} \max_\ell q_\ell T(\ell)^{-1} + O_\infty(\tilde{\eta} N_{\mathrm{trg}} \cdot N^{-1}).
\end{aligned}$$

For all $t \ne \ell_1^* + 2$, observe

$$\begin{aligned}
s_t &\le \tilde{\eta} \frac{q^*}{N_{\mathrm{trg}}} \sum_{w=1}^{N_{\mathrm{trg}}} [3(T^*)^{-1} \cdot 2] + \frac{\tilde{\eta}}{N_{\mathrm{trg}}} [3T(t-2)^{-1} q_{t-2} + 3T(t-3)^{-1} q_{t-3}] + O_\infty(\tilde{\eta} N_{\mathrm{trg}} \cdot N^{-1}) \\
&\le 6\tilde{\eta} \max_\ell q_\ell T(\ell)^{-1} + 6\frac{\tilde{\eta}}{N_{\mathrm{trg}}} \max_\ell q_\ell T(\ell)^{-1} + O_\infty(\tilde{\eta} N_{\mathrm{trg}} \cdot N^{-1}).
\end{aligned}$$

Therefore, we obtain $\frac{1}{2} s_{\ell_1^*+2} \ge s_t$ as desired, if max-sum ratio is $O(N_{\mathrm{trg}}^{-1})$. □

B.4 Proof of Theorem 5

Theorem 5 is directly obtained by combining Lemmas 13 and 15: it only remains to adjust the stepsize.

From Cororally 10 we need $\tilde{\eta}_V = \eta_V / N \lesssim 1/N$, and from (B.5) we need $\tilde{\eta}_V \eta_{KQ} \gtrsim \log N \frac{N^2}{N_{\mathrm{trg}}^3}$. Therefore, it suffices to set

$$\eta_V \lesssim 1 \quad \text{and} \quad \eta_V \eta_{KQ} \gtrsim \frac{N^3}{N_{\mathrm{trg}}^3} \log N.$$

C Finite Sample Analysis

Now we turn to make an analysis for finite sample size setting.

Proof Sketch. We explain how to evaluate the concentration of the empirical gradient $\nabla_{\boldsymbol{W}_V}\hat{\mathcal{L}}(f_{\mathrm{TF}})$. Concentration for $\nabla_{\boldsymbol{W}_{KQ}}\hat{\mathcal{L}}(f_{\mathrm{TF}})$ can be obtained similarly.

Let $\left\{\left\{\boldsymbol{x}_t^{(i)}\right\}_{t=1}^{T(\ell)_i}\right\}_{i=1}^{M_V}$ be M_V i.i.d. sample sequences, and $\left\{\left\{\boldsymbol{x}_t^{(i_{\ell,k})}\right\}_t\right\}_{i_{\ell,k}=1}^{M_V^{\ell,k}}$ be $M_V^{\ell,k}$ i.i.d. sub-samples conditioned on ℓ and $z_{T+1}=k$. Note that $\sum_{\ell,k} M_V^{\ell,k} = M_V$. The empirical gradient $\nabla_{\boldsymbol{W}_V}\hat{\mathcal{L}}(f_{\mathrm{TF}})$ is expressed as

$$\nabla_{\boldsymbol{W}_V}\hat{\mathcal{L}}(f_{\mathrm{TF}}) \simeq \hat{\mathbb{E}}_{\ell\sim\mathcal{D}_\ell}\hat{\mathbb{E}}_{k\sim\mathrm{Unif}[K]}\left[\hat{\boldsymbol{A}}_{\ell,k}\right] + (\text{similar terms omitted})$$

where $\hat{\boldsymbol{A}}_{\ell,k}$ is the empirical average of $\frac{1}{T}\sum_{t=1}^{T}\mathbf{1}_N\boldsymbol{x}_t^{(i_{\ell,k})\top}$ $(i_{\ell,k}=1,\dots,M_V^{\ell,k})$ with the sample size $=M_V^{\ell,k}$. We focus on bounding the first term in this sketch. The gap between the empirical and population gradients is bounded as

$$\begin{aligned}&\|\nabla_{\boldsymbol{W}_V}\hat{\mathcal{L}}(f_{\mathrm{TF}}) - \nabla_{\boldsymbol{W}_V}\mathcal{L}(f_{\mathrm{TF}})\|_2\\ \lesssim&\left\|\hat{\mathbb{E}}_\ell\hat{\mathbb{E}}_k\left[\hat{\boldsymbol{A}}_{\ell,k}-\mathbb{E}[\hat{\boldsymbol{A}}_{\ell,k}|\ell,k]\right]\right\|_2 + \left\|\hat{\mathbb{E}}_\ell\left(\hat{\mathbb{E}}_k-\mathbb{E}_k\right)\mathbb{E}[\hat{\boldsymbol{A}}_{\ell,k}|\ell,k]\right\|_2 + \left\|\left(\hat{\mathbb{E}}_\ell-\mathbb{E}_\ell\right)\mathbb{E}[\hat{\boldsymbol{A}}_{\ell,k}|\ell]\right\|_2.\end{aligned}$$

The first term is bounded by the matrix Hoeffding's inequality using $\|\hat{\boldsymbol{A}}_{\ell,k}\hat{\boldsymbol{A}}_{\ell,k}^\top\|_2, \|\hat{\boldsymbol{A}}_{\ell,k}^\top\hat{\boldsymbol{A}}_{\ell,k}\|_2 \lesssim 1$ and $M_V^{\ell,k}\simeq q_\ell N^{-1}M_V$ for each pair (ℓ,k). Note that $\sum_\ell\sqrt{q_\ell}$ emerges in this bound because $\|\hat{\boldsymbol{A}}_{\ell,k}-\mathbb{E}[\hat{\boldsymbol{A}}_{\ell,k}|\ell,k]\|_2\simeq q_\ell^{-1/2}$ and $\hat{\mathbb{E}}_\ell[\cdot]\simeq\sum_\ell\cdot q_\ell$. The second and third terms can also be bounded by $\|\hat{\boldsymbol{A}}_{\ell,k}\|_2\lesssim 1$ and using the standard Hoeffding's inequality.

C.1 Value Matrix

We first establish an upper-bound for the difference between empirical and population gradient with respect to $\boldsymbol{W}_V$.

Lemma 16. *Let $\boldsymbol{A}_t = \mathbf{1}_{1:N}\boldsymbol{x}_t^\top = \mathbf{1}_{1:N}[\boldsymbol{p}_t^\top\ \boldsymbol{e}_{z_t}^\top\ \boldsymbol{e}_{z_{t-1}}^\top]$. Then we have*

$$\begin{bmatrix}\boldsymbol{A}_t\boldsymbol{A}_t^\top & \mathbf{0}\\ \mathbf{0} & \boldsymbol{A}_t^\top\boldsymbol{A}_t\end{bmatrix}\preceq\boldsymbol{\Sigma}_t$$

where $\lambda_i(\boldsymbol{\Sigma}_t)\lesssim N$ for $i=1,2$ $\lambda_i(\boldsymbol{\Sigma}_t)=0$ for $i\geq 3$ and $\|\boldsymbol{\Sigma}_t\|_2\lesssim N$. Similarly, let $\boldsymbol{B}_{t,k}=\boldsymbol{e}_k\boldsymbol{x}_t^\top$. Then, we have

$$\begin{bmatrix}\boldsymbol{B}_{t,k}\boldsymbol{B}_{t,k}^\top & \mathbf{0}\\ \mathbf{0} & \boldsymbol{B}_{t,k}^\top\boldsymbol{B}_{t,k}\end{bmatrix}\preceq\boldsymbol{\Sigma}_t'$$

where $\|\boldsymbol{\Sigma}_t'\|_2\lesssim 1$.

Proof. We provide the proof for $\boldsymbol{A}_t$. Proof for $\boldsymbol{B}_t$ can be derived in the same vain.

$$\boldsymbol{A}_t\boldsymbol{A}_t^\top = \mathbf{1}_{1:N}\boldsymbol{x}_t^\top\boldsymbol{x}_t\mathbf{1}_{1:N}^\top = 3\begin{bmatrix}1 & \dots & 1\\ \vdots & \ddots & \vdots\\ 1 & \dots & 1\end{bmatrix}\preceq\boldsymbol{\Sigma}_t^{(a)}$$

where $\lambda_1(\boldsymbol{\Sigma}_t^{(a)})=3N$, $\lambda_i(\boldsymbol{\Sigma}_t^{(a)})=0$ for $i\geq 2$.

$$\boldsymbol{A}_t^\top\boldsymbol{A}_t = \boldsymbol{x}_t\mathbf{1}_{1:N}^\top\mathbf{1}_{1:N}\boldsymbol{x}_t^\top = N\boldsymbol{x}_t\boldsymbol{x}_t^\top\preceq\boldsymbol{\Sigma}_t^{(b)}$$

where $\lambda_1(\boldsymbol{\Sigma}_t^{(b)})=3N$, $\lambda_i(\boldsymbol{\Sigma}_t^{(b)})=0$ for $i\geq 2$. Using

$$\mathrm{rank}(\boldsymbol{A}_t\boldsymbol{A}_t^\top)+\mathrm{rank}(\boldsymbol{A}_t^\top\boldsymbol{A}_t)\leq 2$$

and

$$\lambda_1\left(\begin{bmatrix}\boldsymbol{A}_t\boldsymbol{A}_t^\top & \boldsymbol{O}\\ \boldsymbol{O} & \boldsymbol{A}_t^\top\boldsymbol{A}_t\end{bmatrix}\right)\lesssim\max\{\lambda_1(\boldsymbol{A}_t\boldsymbol{A}_t^\top),\lambda_1(\boldsymbol{A}_t^\top\boldsymbol{A}_t)\},$$

we obtain the conclusion. □

Lemma 17. *Let* $\left\{\left\{\boldsymbol{x}_t^{(i)}\right\}_{t=1}^{T_i}\right\}_{i=1}^{M_V}$ *be i.i.d. sample sequences,* $\left\{\left\{\boldsymbol{x}_t^{(i_T)}\right\}_{t=1}^{T}\right\}_{i_T=1}^{M_V^T}$ *be conditionally i.i.d. sub-samples with fixed* T*, and* $\left\{\left\{\boldsymbol{x}_t^{(i_{T,k})}\right\}_{t}\right\}_{i_{T,k}=1}^{M_V^{T,k}}$ *be conditionally i.i.d. sub-samples with fixed* T *and* $z_{T+1}=k$*. Note that* $\sum_{T,k} M_V^{T,k} = \sum_T M_V^T = M_V$*. Then, with probability at least* $1-O(\epsilon)$*,*

$$\lambda_{\max}\left(\nabla_{\boldsymbol{W}_V}\frac{1}{M_V}\sum_{i=1}^{M_V}\mathcal{L}(\boldsymbol{X}_{1:T^{(i)}}^{(i)};\boldsymbol{W}_{KQ},\boldsymbol{W}_V)-\nabla_{\boldsymbol{W}_V}\bar{\mathcal{L}}(\boldsymbol{W}_{KQ},\boldsymbol{W}_V)\right)$$
$$\lesssim\left(\sum_{\ell}\sqrt{q_\ell}\right)\sqrt{\frac{N\log\left(NL(N+L)\epsilon^{-1}\right)}{M_V}}.$$

Proof. The empirical gradient is, noting that $T=2\ell+3\geq 5$,

$$\begin{aligned}\nabla_{\boldsymbol{W}_V}\hat{\mathcal{L}}(f_{\mathrm{TF}}) =&\frac{1}{N}\sum_{k=1}^{N}\boldsymbol{e}_k\hat{\mathbb{E}}_T\left[\frac{1}{T}\sum_{t=1}^{T}\hat{\mathbb{E}}[\boldsymbol{x}_t]^\top\right]\\ &-\sum_{k=N_{\mathrm{trg}}+1}^{N}\boldsymbol{e}_k\hat{\mathbb{E}}_T\left[\frac{1}{T}\sum_{t=1}^{T}\hat{\mathbb{E}}[\boldsymbol{x}_t\mathbb{I}[z_{T+1}=k]]^\top\right]\\ =&\frac{1}{M_V}\sum_{T=5}^{L-1}\sum_{i_T=1}^{M_V^T}\left(\frac{1}{TN}\sum_{t=1}^{T}\mathbf{1}_{1:N}\boldsymbol{x}_t^{(i_T)\top}\right)\\ &-\sum_{T=5}^{L-1}\frac{M_V^T}{M_V}\sum_{k=N_{\mathrm{trg}}+1}^{N}\frac{M_V^{T,k}}{M_V^T}\frac{1}{M_V^{T,k}}\sum_{i_{T,k}=1}^{M_V^{T,k}}\left(\frac{1}{T}\sum_{t=1}^{T}\boldsymbol{e}_k\boldsymbol{x}_t^{(i_{T,k})\top}\right).\end{aligned}$$

Let

$$\boldsymbol{A}^{(i_T)}=\sum_{t=1}^{T}\mathbf{1}_{1:N}\boldsymbol{x}_t^{(i_T)\top}-\mathbb{E}\left[\sum_{t=1}^{T}\mathbf{1}_{1:N}\boldsymbol{x}_t^{(i_T)\top}\mid T\right]$$

and

$$\boldsymbol{B}^{(i_{T,k})}=\sum_{t=1}^{T}\boldsymbol{e}_k\boldsymbol{x}_t^{(i_{T,k})\top}-\mathbb{E}\left[\sum_{t=1}^{T}\boldsymbol{e}_k\boldsymbol{x}^{(i_{T,k})\top}_t\mid y=k,T\right].$$

Using Lemma 16, we can bound the operator norms as

$$\frac{1}{T^2}\left\|\begin{bmatrix}\boldsymbol{A}^{(i_T)}\boldsymbol{A}^{(i_T)\top} & \boldsymbol{O}\\ \boldsymbol{O} & \boldsymbol{A}^{(i_T)\top}\boldsymbol{A}^{(i_T)}\end{bmatrix}\right\|_2\lesssim\sqrt{N}$$

and

$$\frac{1}{T^2}\left\|\begin{bmatrix}\boldsymbol{B}^{(i_{T,k})}\boldsymbol{B}^{(i_{T,k})\top} & \boldsymbol{O}\\ \boldsymbol{O} & \boldsymbol{B}^{(i_{T,k})\top}\boldsymbol{B}^{(i_{T,k})}\end{bmatrix}\right\|_2\lesssim 1.$$

By combining the matrix Hoeffding's inequality and the union bound, we have

$$\left\|\frac{1}{M_V^T TN}\sum_{i_T=1}^{M_V}\boldsymbol{A}^{(i_T)}\right\|_2\lesssim\frac{1}{\sqrt{NM_V^T}}\sqrt{\log\left(L(N+L)\epsilon^{-1}\right)}$$

and

$$\left\|\frac{1}{M_V^{T,k}}\sum_{i_{T,k}=1}^{M_V^{T,k}}\frac{1}{T^{(i_{T,k})^2}}\boldsymbol{B}^{(i_{T,k})}\right\|_2\lesssim\frac{1}{\sqrt{M_V^{T,k}}}\sqrt{\log\left(NL(N+L)\epsilon^{-1}\right)}.$$